4 A hydrogen atom consists of a proton and an electron separated by distance $2x$, as shown in Figure 4.1.

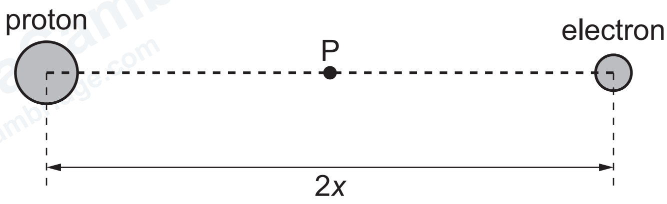

Figure 4.1

Point P is at the midpoint of the line joining the centres of the proton and the electron.

(a) **(i)** For point P, compare the electric field due to the proton with the electric field due to the electron.

....

....

....

....

.... [3]

(ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron.

....

....

....

....

.... [3]

(b) **(i)** The elementary charge is e and the permittivity of free space is ε_0.

State an expression, in terms of x, e and ε_0, for the magnitude of the electric force F between the proton and the electron.

F = ... [2]

(ii) Determine an expression, in terms of F and x, for the electric potential energy E_P of the proton and the electron.

E_P = ... [1]

[Total: 9]